

Locking Away Prosperity? Evaluating Economic Impact of the Key to NYC Program

1 Introduction

The onset of the COVID-19 pandemic triggered a global wave of non-pharmaceutical interventions (NPIs), as governments acted swiftly to limit viral transmission. Early research focused heavily on the public health rationale behind these policies—particularly the importance of acting early to contain exponential spread of the virus. For example, [Friedson et al. \[2021\]](#) and [Yang \[2021\]](#) evaluate the effects of lockdowns in California and Wuhan. Both papers employ synthetic control methods to evaluate the impact of the respective quarantine/shelter in place interventions, emphasizing the critical role of early timing for maximal effect. In a similar note, [Bayat et al. \[2020\]](#) estimate that New York City could have reduced COVID-19 deaths by over 80% had its first lockdown been enacted just ten days earlier.

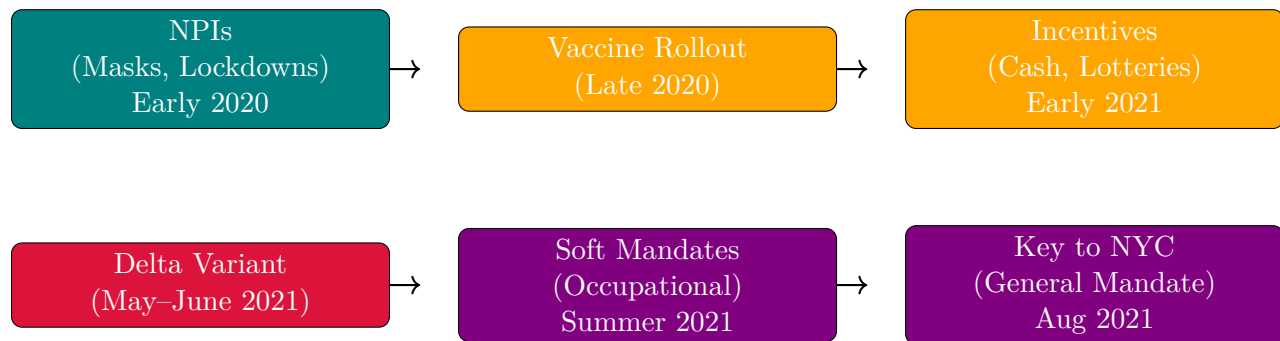
Scholars also paid attention to more secondary outcomes, such as the economic consequences of these policies aside. While many NPIs helped flatten the curve, they also generated medium to longer term concerns about business closures, layoffs, and economic scarring which could persist beyond the policy intervention. Studies such as [Gibson and Sun \[2023\]](#) and [Lee and Yang \[2022\]](#) document the labor market costs of early lockdowns, while [Ke and Hsiao \[2022\]](#) use a panel data approach to examine the short-run economic toll of Hubei’s province-wide shutdown. These papers illustrate the inherent tradeoffs between public health benefits and economic costs embedded in pandemic interventions. In particular, [Ke and Hsiao \[2022\]](#) find that while negative economic impact may happen, the losses are short term when the intervention is sufficiently strict.

Vaccine mandates, unlike lockdowns or school closures, have received comparatively less economic scrutiny, especially regarding their effects on labor markets. These policies aimed to reduce transmission without broadly restricting business operations or movement. Nevertheless, critics warned mandates could hamper fragile economic recoveries by alienating customers and straining staffing levels in consumer-facing sectors like restaurants and hospitality. Proponents countered that mandates would enhance public safety and bolster consumer confidence. New York City’s “Key to NYC” program—the first large-scale vaccine mandate for indoor commercial activity in the U.S., implemented in August 2021—sparked national debate along these lines [[Towey, 2021](#), [Rogers, 2021](#), [Bardosh et al., 2022](#)]. This controversy brings us to the empirical question: *Did the vaccine mandate harm employment levels in NYC’s restaurant sector?* This paper treats the Key to NYC program as a natural experiment,

exploiting its rollout to identify its causal impact on full-service restaurant employment—the segment most directly affected by vaccine mandates. While prior research has focused on mandates’ effects on health outcomes and vaccination rates [Cohn et al., 2022, Acharya and Dhakal, 2021, Lang et al., 2023], few have rigorously examined labor market consequences in the places most likely to be affected by them. This study is among the first to isolate its economic impact on the full-service restaurant sector—the area arguably most exposed to the policy’s constraints.

Randomized experiments are infeasible and unethical for assigning vaccine mandates to cities, so we must estimate counterfactuals using observational data. These counterfactuals can never be directly observed, but they can be approximated. I construct a synthetic control for the year-over-year growth rate of full-service restaurant employment in New York City using seasonally adjusted MSA-level data from the Federal Reserve Bank of St. Louis. The dataset spans January 1991 to August 2022, allowing for a long pre-treatment window. The extended time series is particularly valuable given the properties of the synthetic control method, which is most reliable in the fixed- N , large- T setting, given that the bound on the bias becomes tighter as the length of the pre-treatment time series increases. There are 29 donor units that FRED reports on to construct the counterfactual from.¹ As a robustness check, I also examine employment in the broader Leisure and Hospitality sector, which may capture indirect spillover effects or broader sectoral shocks correlated with restaurant trends.

2 Policy Background



Timeline of major pandemic policy phases

Building on early reliance on non-pharmaceutical interventions (NPIs) such as lockdowns and masking, the arrival of vaccines in late 2020 ushered in a new phase of pandemic governance. NPIs had aimed to reduce transmission by limiting contact or promoting respiratory barriers. Mask mandates were widely adopted due to their low cost and direct mechanism of action [Hansen and Mano, 2023, Chernozhukov et al., 2021, Mitze et al., 2020], while lockdowns—though effective—provoked political and economic backlash due to their disruptive nature.

¹For example, one such series is ‘All Employees: Leisure and Hospitality: Full-Service Restaurants in District of Columbia (SMU11000007072251101SA)’ or ‘SMU08197407072251101SA’ for Denver-Aurora-Centennial, CO (MSA). These cities did not implement similar mandates during the same period, and thus serve as a “donor pool” for constructing the counterfactual.

With vaccines came the promise of exit, and initial uptake was high, particularly among older adults and healthcare workers. However, as eligibility expanded in early 2021, demand slowed amid misinformation, mistrust, and logistical barriers. In response, policymakers deployed incentive-based strategies to encourage vaccination [Barber and West, 2022, Robertson et al., 2021]. These included direct cash payments, vaccine lotteries (e.g., Ohio’s “Vax-a-Million” [Fuller et al., 2022]), and smaller symbolic giveaways—ranging from donuts and beer to cannabis and firearms. One paper famously dubbed this the era of “donuts, drugs, booze, and guns,” reflecting the extraordinary lengths to which governments went to boost uptake through rewards rather than requirements [Tinari and Riva, 2021].

Despite their creativity, most incentives had modest or short-lived effects [Saban et al., 2021, Walkey et al., 2021, Khazanov et al., 2023]. Many were poorly targeted, administratively complex, or insufficient to overcome deeper resistance. The emergence of the highly transmissible Delta variant in mid-2021 raised the stakes. Initially identified in India, Delta reached the United States by May and accounted for the majority of new cases by June. Unlike in 2020, officials now had vaccines at scale—but not enough vaccinated people.

Public health responses escalated. Soft mandates emerged, targeting workers in high-risk occupations such as healthcare and education. These policies were not general mandates but did make vaccination a condition of continued employment [Khuntia et al., 2021]. Over the summer, local governments expanded this approach. By August 2021, New York City introduced the Key to NYC program—one of the first large-scale vaccine mandates in the U.S. applied to the general public. Unlike prior efforts that relied on encouragement, Key to NYC made access to indoor dining, fitness, and entertainment conditional on vaccination status. It represented a new policy mechanism: exclusion rather than incentive.

3 Dataset

We use monthly MSA-level employment series from the Federal Reserve Bank of St. Louis (FRED). The sample runs from January 1991 through August 2022, yielding $T = 380$ monthly observations per unit. Units are indexed by $j \in \mathbb{N}$ and time by $t \in \mathbb{N}$. Unit $j = 1$ is the treated unit (New York City MSA). The donor pool is $\mathcal{N}_0 = \mathcal{N} \setminus \{1\}$ with cardinality N_0 ; in the baseline sample $N_0 = 29$ (so $N = 30$ total MSAs).

Let e_{jt} denote employment in full-service restaurants in unit j at month t . The primary outcome is the year-over-year growth rate (in percentage points),

$$y_{jt} = 100 \cdot \frac{e_{jt} - e_{j,t-12}}{e_{j,t-12}}, \quad j = 1, \dots, N, \quad t = 13, \dots, T.$$

Collect outcomes into vectors $\mathbf{y}_j = (y_{j1}, \dots, y_{jT})^\top \in \mathbb{R}^T$ and the donor matrix

$$\mathbf{Y}_0 = [\mathbf{y}_j]_{j \in \mathcal{N}_0} \in \mathbb{R}^{T \times N_0}.$$

The *Key to NYC* policy was implemented in August 2021. Let T_0 denote the last pre-treatment month (July 2021). Define the pre- and post-treatment index sets $\mathcal{T}_1 = \{t \leq T_0\}$

and $\mathcal{T}_2 = \{t > T_0\}$. For convenience we partition vectors and matrices into pre/post blocks, e.g. $\mathbf{y}_j = (\mathbf{y}_{j,\mathcal{T}_1}^\top, \mathbf{y}_{j,\mathcal{T}_2}^\top)^\top$.

Let $y_{jt}(0)$ and $y_{jt}(1)$ denote the potential outcomes for unit j at time t under no treatment and treatment, respectively. Define the observed treatment indicator

$$d_{jt} = \mathbf{1}\{j = 1 \wedge t > T_0\},$$

so only the treated unit is exposed in the post period. The observed outcome satisfies the switching equation (fundamental problem of causal inference):

$$y_{jt} = (1 - d_{jt}) y_{jt}(0) + d_{jt} y_{jt}(1).$$

Consequently only one of $\{y_{jt}(0), y_{jt}(1)\}$ is observed for each (j, t) , and the causal effect for unit j at time t is

$$\tau_{jt} = y_{jt}(1) - y_{jt}(0).$$

Our primary object of interest is the post-treatment average treatment effect on the treated (ATT) for the treated unit $j = 1$:

$$\text{ATT} = \frac{1}{|\mathcal{T}_2|} \sum_{t \in \mathcal{T}_2} [y_{1t}(1) - y_{1t}(0)].$$

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